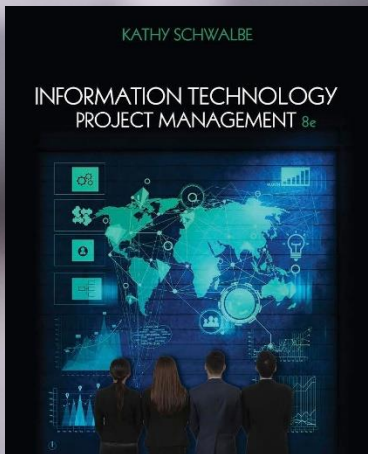




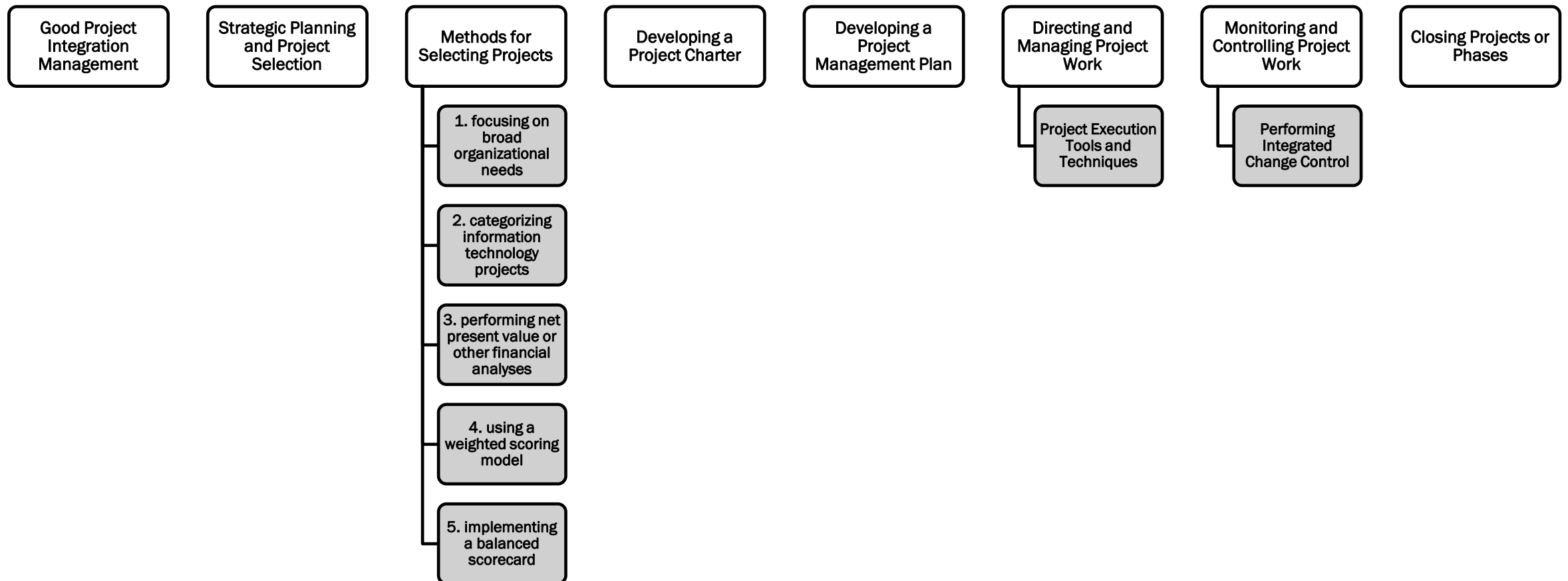
Note:

Adapted from slide of the textbook: Schwalbe, Kathy. Managing Information Technology Project – Eight Edition. Boston, MA: Thomson Course Technology, 2016.
See the text itself for full citations.

IT PROJECT MANAGEMENT

WEEK 4 : PROJECT INTEGRATION
MANAGEMENT

MIND MAP



LEARNING OBJECTIVES

PART 1

- a) Describe an **overall framework for project integration** management as it relates to the other project management knowledge areas and the project life cycle
- b) Discuss **the strategic planning process** and apply different project **selection methods**

PART 2

- a) Explain the importance of creating a **project charter** to formally initiate projects
- b) Describe **project management plan** development, understand the content of these plans, and review approaches for creating them
- c) Explain **project execution**, its relationship to project planning, the factors related to successful results, and tools and techniques to assist in directing and managing project work
- d) Describe the **process of monitoring and controlling** a project
 - Understand the **integrated change control process**, planning for and managing changes on information technology (IT) projects, and developing and using a change control system
- e) Explain the importance of developing and following good procedures for **closing projects**



PART 1.A: WHAT IS PROJECT INTEGRATION MANAGEMENT?



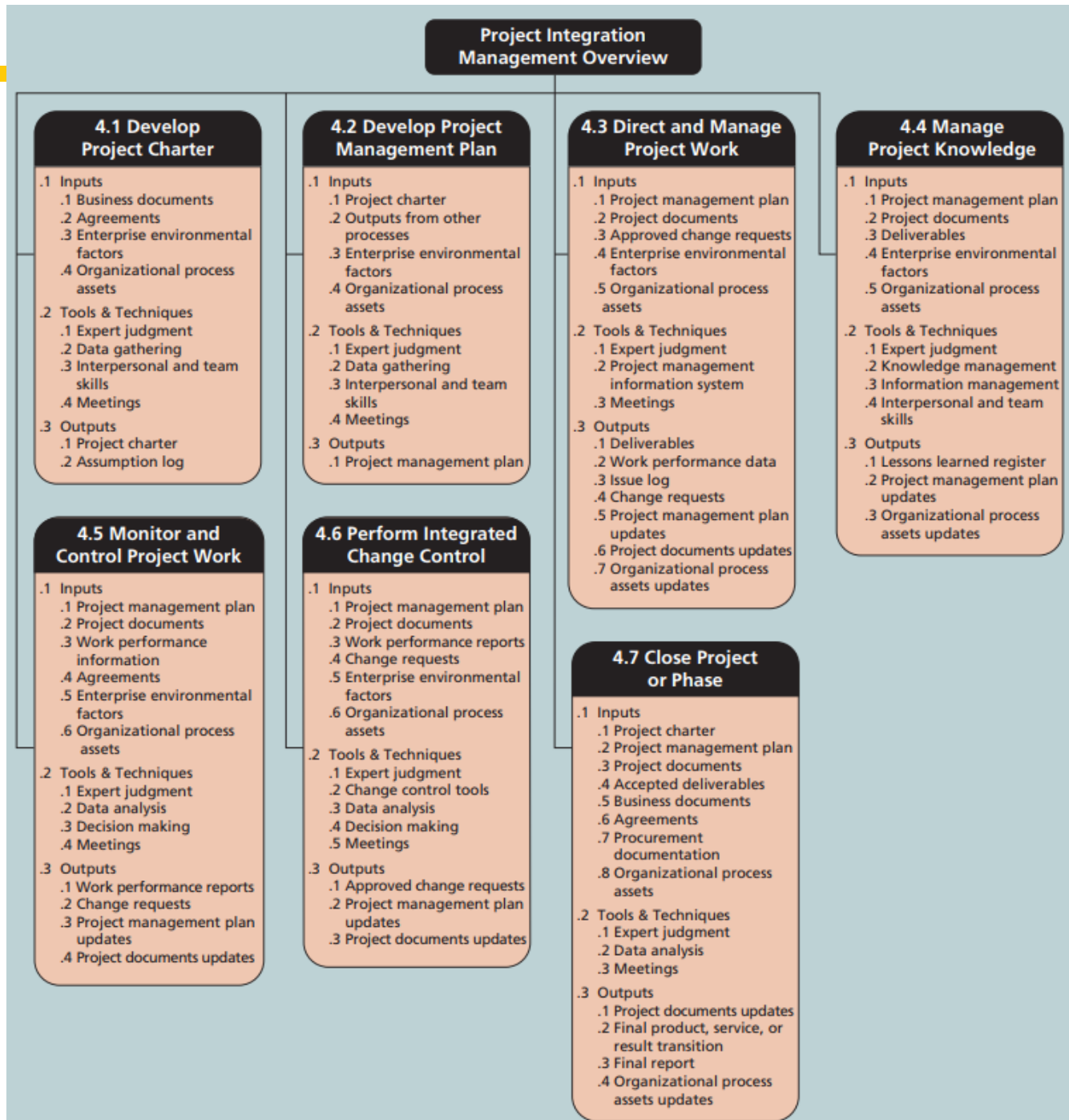
THE KEY TO OVERALL PROJECT SUCCESS: GOOD PROJECT INTEGRATION MANAGEMENT

- **Project integration management** involves coordinating all of the other project management knowledge areas throughout a project's life cycle.
- Project managers must coordinate all of the other knowledge areas throughout a project's life cycle. Project Manager must take responsibility for coordinating all of the people, plans, and work required to complete a project
- Many new project managers have trouble looking at the “big picture” and want to focus on too many details
- Project integration management is not the same thing as software integration

PROJECT INTEGRATION MANAGEMENT PROCESSES

- 1) **Developing the project charter** involves working with stakeholders to create the document that formally authorizes a project—the charter.
- 2) **Developing the project management plan** involves coordinating all planning efforts to create a consistent, coherent document—the project management plan.
- 3) **Directing and managing project work** involves carrying out the project management plan by performing the activities included in it.
- 4) **Managing project knowledge** involves using existing knowledge and creating new knowledge to achieve project objectives while also contributing to organizational learning
- 5) **Monitoring and controlling** project work involves overseeing activities to meet the performance objectives of the project
- 6) **Performing integrated change control** involves identifying, evaluating, and managing changes throughout the project life cycle.
- 7) **Closing the project or phase** involves finalizing all activities to formally close the project or phase.

PROJECT INTEGRATION MANAGEMENT SUMMARY





PART 1.B: STRATEGIC PLANNING AND PROJECT SELECTION

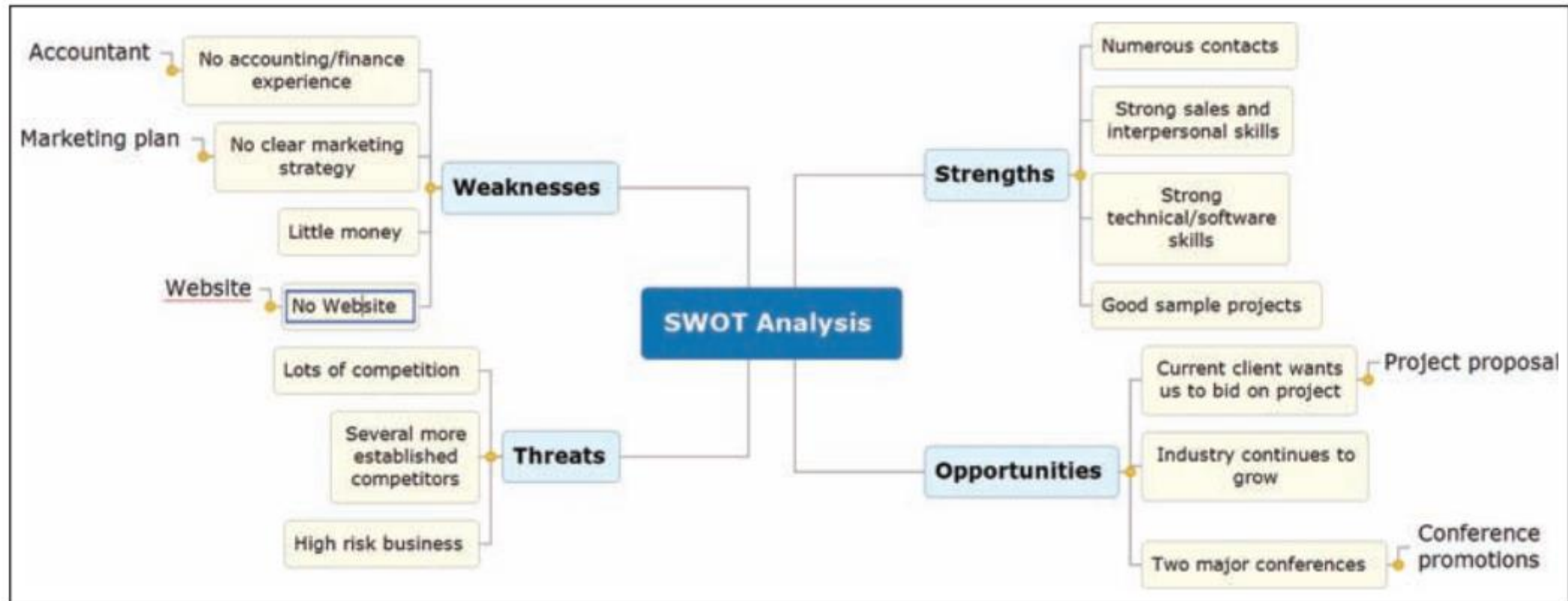


STRATEGIC PLANNING AND PROJECT SELECTION

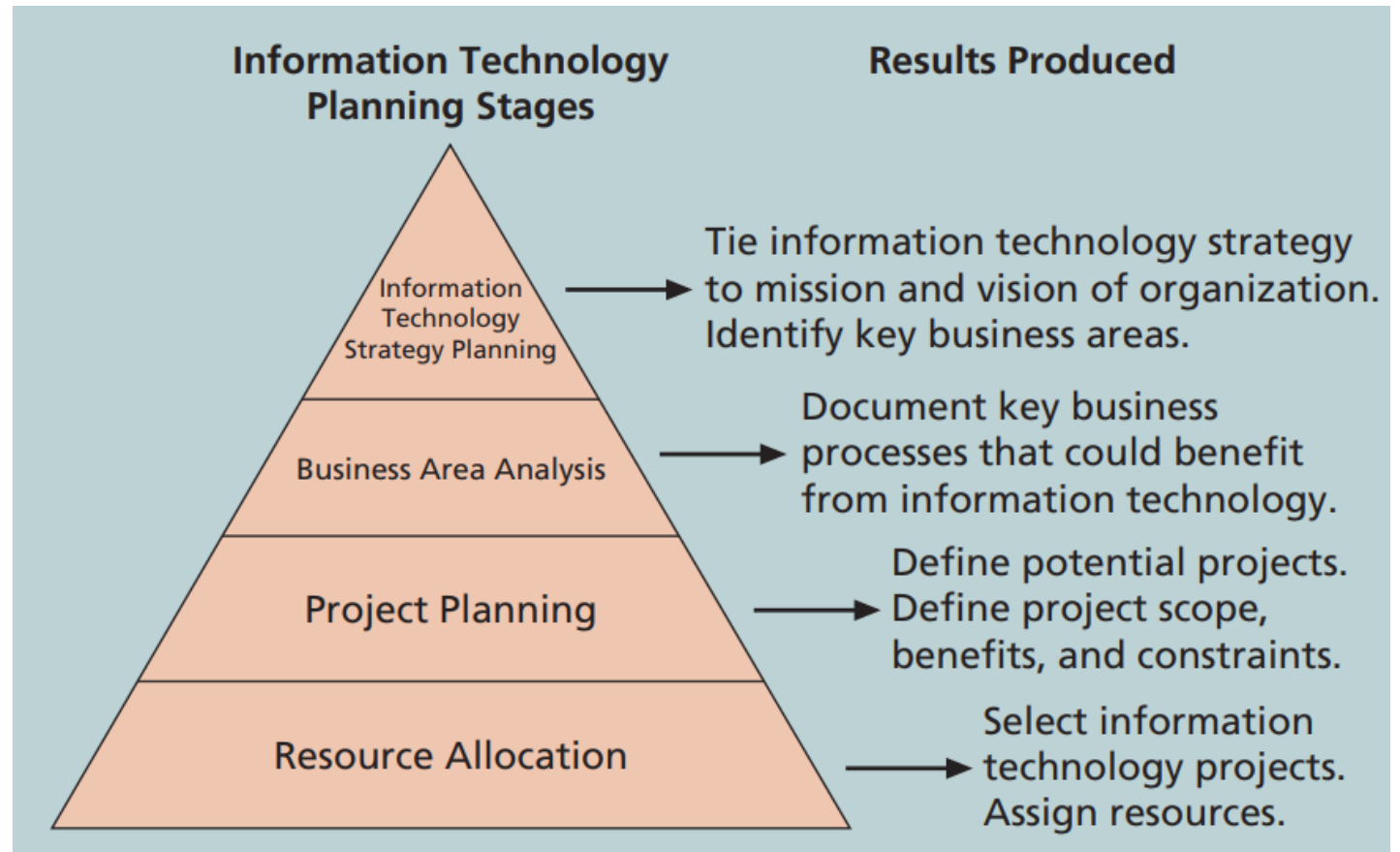
- **Strategic planning** involves determining long-term objectives, predicting future trends, and projecting the need for new products and services
- Organizations often perform a **SWOT analysis** to analyze **S**trengths, **W**eaknesses, **O**pportunities, and **T**hreats
- As part of strategic planning, organizations
 - **identify** potential projects
 - use realistic methods to **select** which projects to work on
 - **formalize** project initiation by issuing a project charter

MIND MAP OF A SWOT ANALYSIS TO HELP IDENTIFY POTENTIAL PROJECTS

For example, a group of four people who want to start a new business in the film industry could perform a SWOT analysis to help identify potential projects.



INFORMATION TECHNOLOGY PLANNING PROCESS



ALIGNING IT WITH BUSINESS STRATEGY

- Aligning IT projects with business strategy is at the heart of selecting IT projects
- An organization must develop a strategy for using IT to define how it will **support the organization's objectives**. IT strategy must align with the organization's strategic plans.
- Research shows that supporting explicit business objectives is the top reason cited for **why organizations invest in IT projects**. Other top criteria for investing in IT projects include supporting implicit business objectives and providing financial incentives, such as a good internal rate of return (IRR) or net present value (NPV)
- Using a balanced scorecard approach, as described later in this chapter also helps align projects with business strategy.
- The majority of survey respondents felt that the **CIO role was becoming more strategic**, reporting that 62 percent sit on the executive board, an all-time high from past surveys



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METHODS FOR SELECTING PROJECTS

- There are usually more projects than available time and resources to implement them.
- Methods for selecting projects include:
 1. focusing on broad organizational needs
 2. categorizing information technology projects
 3. performing net present value or other financial analyses
 4. using a weighted scoring model
 5. implementing a balanced scorecard
- In practice, many organizations use a combination of these approaches to select projects

1. FOCUSING ON BROAD ORGANIZATIONAL NEEDS

- It is often difficult to provide strong justification for many IT projects, but everyone agrees they have a high value
- “It is better to measure gold roughly than to count pennies precisely”
- Three important criteria for projects:
 - There is a ***need*** for the project
 - There are ***funds*** available
 - There’s a strong ***will*** to make the project succeed

2. CATEGORIZING IT PROJECTS

- One categorization is whether the project addresses
 - a problem
 - an opportunity, or
 - a directive
- Another categorization is how long it will take to do and when it is needed
- Another is the overall priority of the project

3. FINANCIAL ANALYSIS OF PROJECTS

- Financial considerations are often an important consideration in selecting projects
- Three primary methods for determining the projected financial value of projects:
 - Net present value (NPV) analysis
 - Return on investment (ROI)
 - Payback analysis

NET PRESENT VALUE ANALYSIS

- **Net present value (NPV)** analysis is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time
- Projects with a **positive NPV** should be considered if financial value is a key criterion
- The higher the NPV, the better

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	\$0	\$2,000	\$3,000	\$4,000	\$5,000	\$14,000
5	Costs	\$5,000	\$1,000	\$1,000	\$1,000	\$1,000	\$9,000
6	Cash flow	(\$5,000)	\$1,000	\$2,000	\$3,000	\$4,000	\$5,000
7	NPV →	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV →	\$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money

NPV CALCULATIONS

- Determine estimated costs and benefits for the life of the project and the products it produces
- Determine the discount rate (check with your organization on what to use)
- Calculate the NPV

$$NPV = \sum_{t=0 \dots n} A_t / (1 + r)^t$$

- t equals the year of the cash flow
- n is the last year of the cash flow
- A is the amount of cash flow each year
- r is the discount rate

Notes: Some organizations consider the investment year as year 0, while others start in year 1. Some people entered costs as negative numbers, while others do not. Check with your organization for their preferences

The formula for the discount factor is

$$\text{Discount factor} = 1 / (1 + r)^t$$

where r is the discount rate, and t is the year

JWD CONSULTING NPV EXAMPLE

Discount rate	8%					
Assume the project is completed in Year 0			Year			
	0	1	2	3	Total	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	140,000	37,200	34,400	31,600	243,200	
Benefits	0	200,000	200,000	200,000		
Discount factor	1	0.93	0.86	0.79		
Discounted benefits	0	186,000	172,000	158,000	516,000	
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
ROI	112%					
	Payback in Year 1					

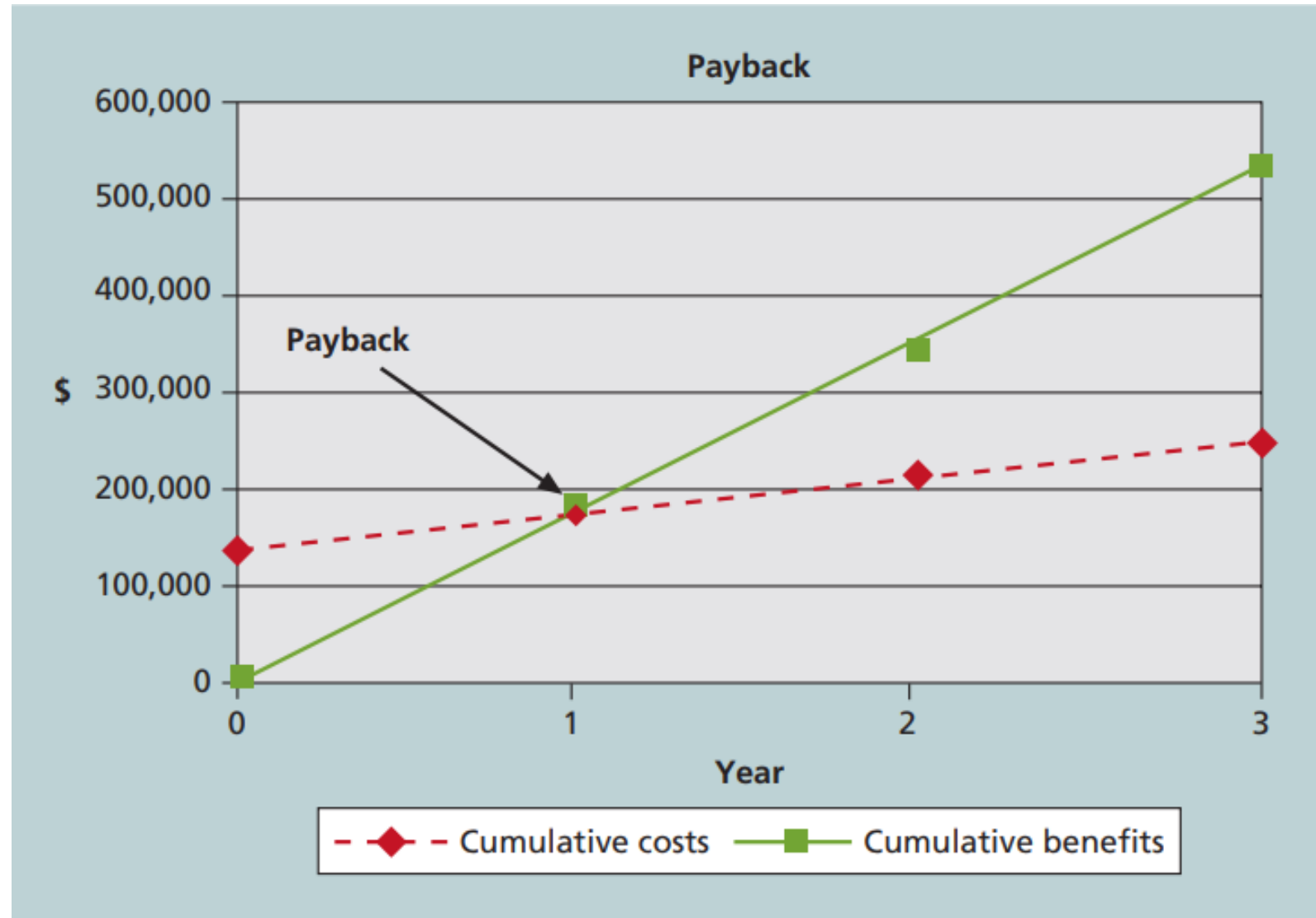
RETURN ON INVESTMENT

- **Return on investment (ROI)** is calculated by subtracting the project costs from the benefits and then dividing by the costs
$$\text{ROI} = (\text{total discounted benefits} - \text{total discounted costs}) / \text{discounted costs}$$
- The higher the ROI, the better
- Many organizations have a **required rate of return** or minimum acceptable rate of return on investment for projects
- **Internal rate of return (IRR)** can be calculated by finding the discount rate that makes the NPV equal to zero

PAYBACK ANALYSIS

- Another important financial consideration is payback analysis
- The **payback period** is the amount of time it will take to recoup, in the form of net cash inflows, the total dollars invested in a project
- Payback occurs when the net cumulative discounted benefits equals the costs
- Many organizations want IT projects to have a fairly short payback period

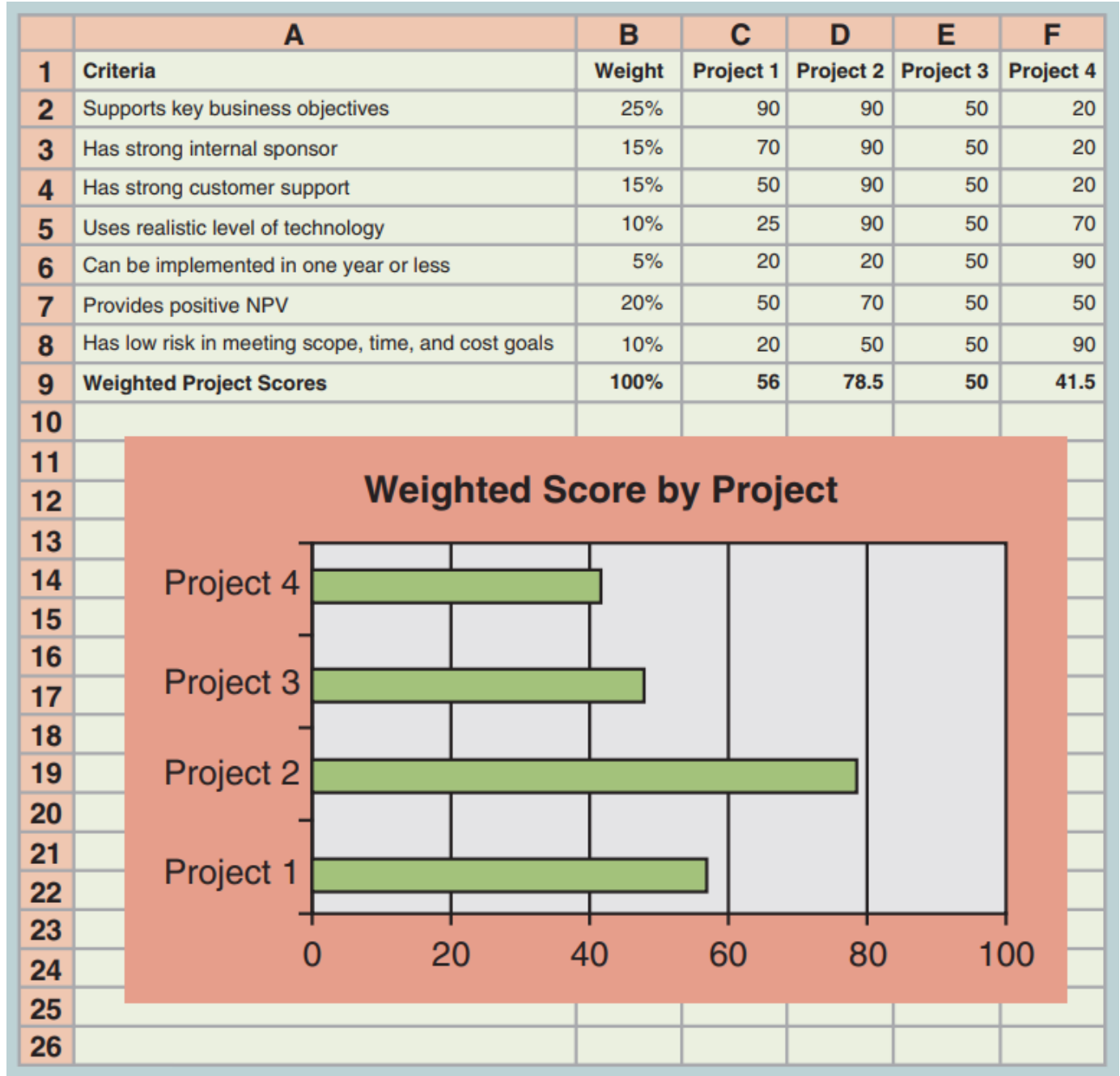
CHARTING THE PAYBACK PERIOD



4. WEIGHTED SCORING MODEL

- A weighted scoring model is a tool that provides a systematic process for selecting projects based on many criteria
 - Identify criteria important to the project selection process
 - Assign weights (percentages) to each criterion so they add up to 100%
 - Assign scores to each criterion for each project
 - Multiply the scores by the weights and get the total weighted scores
- The higher the weighted score, the better

SAMPLE WEIGHTED SCORING MODEL FOR PROJECT SELECTION



5. IMPLEMENTING A BALANCED SCORECARD

- Drs. Robert Kaplan and David Norton developed this approach to help select and manage projects that align with business strategy
- A **balanced scorecard**
 - is a methodology that converts an organization's value drivers, such as customer service, innovation, operational efficiency, and financial performance, to a series of defined metrics
- See www.balancedscorecard.org for more information



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PART 2.A: DEVELOPING A PROJECT CHARTER



DEVELOPING A PROJECT CHARTER

- After deciding what project to work on, it is important to let the rest of the organization know
- A **project charter** is a document that **formally** recognizes the **existence** of a project and provides direction on the project's objectives and management
- Key project stakeholders should **sign** a project charter to acknowledge agreement on the need and intent of the project; **a signed charter is a key output of project integration management**
- **Input for developing project charter:**
 - A project statement of work
 - A business case
 - Agreements
 - Enterprise environmental factors
 - **Organizational process assets**, which include formal and informal plans, policies, procedures, guidelines, information systems, financial systems, management systems, lessons learned, and historical information

PROJECT CHARTER EXAMPLE

Project Title: Next-gen DNA-Sequencing Instrument Completion Project

Date of Authorization: February 1

Project Start Date: February 1 **Projected Finish Date:** November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated \$1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, nearson@dnaconsulting.com

Project Objectives: The Next-gen DNA-sequencing instrument project has been under way for three years. It is a crucial project for our company. This is the first charter for the project; the objective is to complete the first version of the instrument software in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

PROJECT CHARTER EXAMPLE

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the Next-gen DNA-sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA Expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing Expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com

Sign-off: (signatures of all the above stakeholders)

Ahmed Abrams

Nick Carson

Susan Johnson

Renyong Chi

Erik Haus

Bill Strom

Maggie Elliot

Comments: (handwritten or typed comments from above stakeholders, if applicable)

"I want to be heavily involved in this project. It is crucial to our company's success, and I expect everyone to help make it succeed."—ahmed abrams



PART 2.B: DEVELOPING A PROJECT MANAGEMENT PLAN



DEVELOPING A PROJECT MANAGEMENT PLAN

- A **project management plan** is a document used to **coordinate** all project planning documents and help **guide** a project's **execution** and **control**
- Plans created in the other knowledge areas are subsidiary parts of the overall project management plan.
- *Common element in project management plan:*
 - Introduction or overview of the project
 - Description of how the project is organized
 - Management and technical processes used on the project
 - Work to be done, schedule, and budget information
 - References to other project planning documents

SAMPLE CONTENTS FOR A SOFTWARE PROJECT MANAGEMENT PLAN (SPMP)

Major Section Headings	Section Topics
Overview	Purpose, scope, and objectives; assumptions and constraints; project deliverables; schedule and budget summary; evolution of the plan
Project Organization	External interfaces; internal structure; roles and responsibilities
Managerial Process Plan	Start-up plans (estimation, staffing, resource acquisition, and project staff training plans); work plan (work activities, schedule, resource, and budget allocation); control plan; risk management plan; closeout plan
Technical Process Plans	Process model; methods, tools, and techniques; infrastructure plan; product acceptance plan
Supporting Process Plans	Configuration management plan; verification and validation plan; documentation plan; quality assurance plan; reviews and audits; problem resolution plan; subcontractor management plan; process improvement plan

Source: IEEE Standard 1058–1998.



PART 2.C: DIRECTING AND MANAGING PROJECT WORK



DIRECTING AND MANAGING PROJECT WORK

- Involves managing and performing the work described in the project management plan
- The **majority of time and money is usually spent on execution**
- The application area of the project directly affects project execution because the products of the project are produced during execution
- The project manager needs to focus on leading the project team and managing stakeholder relationships to execute the project management plan successfully

COORDINATING PLANNING AND EXECUTION

- Project planning and execution are intertwined and inseparable activities
- Those who will do the work should help to plan the work
- Project managers must solicit input from the team to develop realistic plans
 - Those who will do the work should plan the work

PROVIDING LEADERSHIP AND A SUPPORTIVE CULTURE

- Project managers must lead by example to demonstrate the importance of creating and then following good project plans
- Organizational **culture** can help project execution by
 - providing **guidelines** and **templates**
 - **tracking** performance based on plans
- Project managers may still need to break the rules to meet project goals, and senior managers must support those actions

CAPITALIZING ON PRODUCT, BUSINESS, AND APPLICATION AREA KNOWLEDGE

- It is often helpful for IT project managers to have **prior technical** experience
- On small projects, the project manager may be required to **perform** some of the **technical** work or mentor team members to complete the projects
- On large projects, the project manager must understand the **business** and **application area** of the project

PROJECT EXECUTION TOOLS AND TECHNIQUES

- **Expert judgment:** Experts can help project managers and their teams make many decisions related to project execution
- **Meetings:** Meetings allow people to develop relationships, pick up on important body language or tone of voice, and have a dialogue to help resolve problems.
- **Project management information systems:** There are hundreds of project management software products available on the market today, and many organizations are moving toward powerful enterprise project management systems that are accessible via the Internet

MANAGING PROJECT KNOWLEDGE

- There are two basic types of knowledge:
 - 1. **Explicit knowledge:** This type of knowledge can be easily explained using words, pictures, or numbers and is easy to communicate, store, and distribute. Examples include information found in textbooks and encyclopedias as well as project documents and plans.
 - 2. **Tacit knowledge:** Unlike explicit knowledge, tacit knowledge, sometimes called informal knowledge, is difficult to express and is highly personal. Examples include beliefs, insight, and experience. It is often shared through conversations and interactions between people. Many organizations set up programs like mentorships, communities of practice, or workshops to assist in passing on tacit knowledge.
- Knowledge management should be done before, during, and after projects are completed
- A lessons-learned register should document challenges, problems, realized risks and opportunities, and other content to assist in knowledge management on current and future projects



PART 2.D: MONITORING AND CONTROLLING PROJECT WORK



MONITORING AND CONTROLLING PROJECT WORK

- **Changes are inevitable** on most projects, so it's important to develop and follow a process to monitor and control changes
- Monitoring project work includes collecting, measuring, and disseminating performance information
- A **baseline** is the approved project management plan plus approved changes
- Another critical factor in change control is **communication**.
 - Project managers should use written and oral performance reports to help identify and manage project changes



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PERFORMING INTEGRATED CHANGE CONTROL

- Three main objectives are:
 - Influencing the factors that create changes to ensure that changes are beneficial
 - Determining that a change has occurred
 - Managing actual changes as they occur

Former View

- The project team should strive to do exactly what was planned on time and within budget
- **Problem:** Stakeholders rarely agreed up-front on the project scope, and time and cost estimates were inaccurate

Modern View

- Project management is a process of constant communication and negotiation
- **Solution:** Changes are often beneficial, and the project team should plan for them

CHANGE CONTROL SYSTEM

- A **change control system** is a formal, documented process that describes when and how official project documents and work may be changed
- Describes who is authorized to make changes and how to make them
- **Change Control Board (CCB)**
 - A **change control board** is a formal group of people responsible for approving or rejecting changes on a project
 - CCBs provide guidelines for preparing change requests, evaluate change requests, and manage the implementation of approved changes
 - Includes stakeholders from the entire organization

MAKING TIMELY CHANGES

- Some CCBs only meet occasionally, so it may take too long for changes to occur
- Some organizations have policies in place for time-sensitive changes
 - “48-hour policy” allows project team members to make decisions, then they have 48 hours to reverse the decision pending senior management approval
 - Delegate changes to the lowest level possible, but keep everyone informed of changes



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GLOBAL ISSUES

- Rapid changes in technology, such as the increased use of mobile roaming for communications, often cause governments around the world to take action.
- Incompatible hardware, software, and networks can make communications difficult in some regions, and a lack of competition can cause prices to soar.
- Fortunately, a group called the Organisation for Economic Co-operation and Development (OECD) promotes policies that will improve the economic and social well-being of people around the world.
- In February 2012, the OECD called upon its members' governments to boost competition in international mobile roaming markets.
- By the end of 2013, wireless broadband penetration grew to 72.4% in the OECD area

CONFIGURATION MANAGEMENT

- Configuration management is another important part of integrated change control
- **Configuration management** ensures that the descriptions of the project's products are correct and complete
- Involves **identifying** and **controlling** the **functional** and **physical design** characteristics of products and their support documentation
- Configuration management specialists identify and document configuration requirements, control changes, record and report changes, and audit the products to verify conformance to requirements

SUGGESTIONS FOR PERFORMING INTEGRATED CHANGE CONTROL

View project management as a process of constant communication and negotiation.

Plan for change.

Establish a formal change control system, including a change control board (CCB) and IT steering committee.

Use effective configuration management.

Define procedures for making timely decisions about smaller changes.

Use written and oral performance reports to help identify and manage change.

Use project management software and other software to help manage and communicate changes.

Focus on leading the project team and meeting overall project goals and expectations.



PART 2.E: CLOSING PROJECT OR PHASES



CLOSING PROJECTS OR PHASES

- To close a project or phase, you must **finalize all activities and transfer the completed or cancelled work** to the appropriate people
- The main **inputs** to this process are the project charter, project management plan, project documents, accepted deliverables, business documents, agreements, procurement documentation, and organizational process assets
- The **outputs** of closing projects are as follows :
 - *Project documents updates*: All project documents should be reviewed and marked as final versions
 - *Final product, service, or result transition*: Project sponsors are usually most interested in making sure they receive delivery of the final products, services, or results they expected when they authorized the project
 - *Final report*: A final project report and presentation are also commonly used during project closing
 - *Organizational process asset updates*: The project team should provide a list of project documentation, project closure documents, and historical information produced by the project in a useful format.

CHAPTER SUMMARY

- Project integration management involves coordinating all of the other knowledge areas throughout a project's life cycle
- Main processes include
 - Develop the project charter
 - Develop the project management plan
 - Direct and manage project execution
 - Monitor and control project work
 - Perform integrated change control
 - Close the project or phase